

Parth Mishra

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LeetCode: leetcode.com/u/parth213g
Location: Bhopal, India (Open to Pan-India & Remote)

SKILLS SUMMARY

- **Languages:** C++ (Proficient), Python (Proficient), JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
- **Frameworks & Libraries:** React, React Native, Node.js, Express.js, FastAPI, Pydantic, Flask, Tailwind CSS, Vite
- **AI & Systems Engineering:** Agentic Systems, RAG Pipelines, Vector Search (HNSW, KD-Trees), Search (A*, Minimax), Bayesian Inference, Claude API, Ollama
- **Tools, Cloud & Databases:** PostgreSQL, MySQL, Docker, Git, GitHub, Render, Postman, Linux CLI, RESTful APIs
- **Core Computer Science:** Data Structures & Algorithms (DSA), OOP, DBMS, Operating Systems, Computer Networks
- **Engineering Practices:** Clean Architecture, Distributed Systems, Unit & Integration Testing (PyTest, Jest), CI/CD Workflows, Agile Sprints

EDUCATION

- **Vellore Institute of Technology (VIT Bhopal)** **Bhopal, India**
Bachelor of Technology – Computer Science and Engineering (AI/ML); CGPA: 8.67 *August 2023 – May 2027*
Core Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, DBMS, OS, Computer Networks, NLP

EXPERIENCE

- **Craftedge Academy** **Remote / Bhopal, India**
Software Engineering Intern (Front-End & Systems) *October 2025 – April 2026*
 - **Production UI Engineering:** Architected and deployed modular, responsive web components in React and JavaScript across client-facing academy platforms serving 1,000+ active learners, achieving 100% QA verification.
 - **Performance & Latency Optimization:** Reduced initial page load latency by 28% through aggressive component code-splitting, lazy-loading, and asset minification; eliminated layout shifts across mobile viewports.
 - **State Management & Zero Regressions:** Built centralized client-side state workflows and debounced REST API queries, cutting redundant network requests by 35% with zero regressions across release cycles.

PROJECTS

- **VectorDB – In-Memory Vector Search Engine & RAG Pipeline:** Engineered an in-memory vector database from first principles in Python implementing Hierarchical Navigable Small World (HNSW) graph traversal and KD-Trees. Achieved sub-15ms nearest-neighbor retrieval over 50,000+ 768-dim embeddings, delivering a 38x speedup over brute-force search. Developed an end-to-end RAG pipeline chunking documents and serving grounded context to local Ollama LLMs.
Tech: Python, Flask, HNSW Indexing, KD-Trees, Ollama, REST APIs [GitHub]
- **Insta Insights – Mobile Analytics Platform & Licensing Engine:** Architected a full-stack mobile analytics and license-key authorization platform in React Native, Node.js, and Express, deployed on Render. Designed normalized PostgreSQL schemas with connection pooling and ACID-compliant transaction boundaries; engineered secure JWT-based authentication and rate-limited REST endpoints maintaining sub-80ms API response latency.
Tech: React Native, Node.js, Express.js, PostgreSQL, Render, REST APIs [GitHub]
- **Autonomous Career & Placement Workflow Engine:** Architected an event-driven Python automation system for multi-portal discovery, applicant tracking, and semantic ATS keyword alignment. Built an asynchronous IMAP monitoring daemon parsing campus placement notices and deadlines in real time, integrated with automated document compilers and verified SMTP dispatch.
Tech: Python, BeautifulSoup, IMAP/SMTP Protocols, XeLaTeX, Systems Automation, REST APIs [GitHub]

ACHIEVEMENTS

- **Competitive Programming & Problem Solving:** Solved 250+ algorithmic problems across LeetCode, Geeks-for-Geeks, and HackerRank; established pattern mastery in Dynamic Programming, Graphs, Trees, and Two Pointers.
- **Academic Distinction (CGPA 8.67):** Maintained top percentile academic standing in Computer Science & Engineering (AI/ML) across all evaluated semesters at VIT Bhopal.
- **Harvard CS50 AI Capstone Distinction:** Awarded verified credential upon implementing 12 foundational AI systems from scratch (Minimax with Alpha-Beta Pruning, HMMs, Neural Networks, Attention models) with 100% test compliance.

CERTIFICATIONS

- **CS50's Introduction to Artificial Intelligence with Python – Harvard University** **Oct 2026**
12-Project AI Mastery: Search, Knowledge, Logic, Probability, ML, Neural Networks & Transformers [Verify: 3e7e245a]
- **Oracle Certified Foundations Associate – Agentic AI – Oracle University** **Sept 2026**
Autonomous Agent Architectures, Multi-Agent Systems, Tool Integration & Enterprise LLMs [Verify ID: 103546830AAI26OFA]
- **Applied Machine Learning in Python – University of Michigan** **Jan 2026**
Supervised & Unsupervised Learning, Feature Engineering, Scikit-Learn Model Pipelines [Verify: G1Q6ZYPKPWP4]